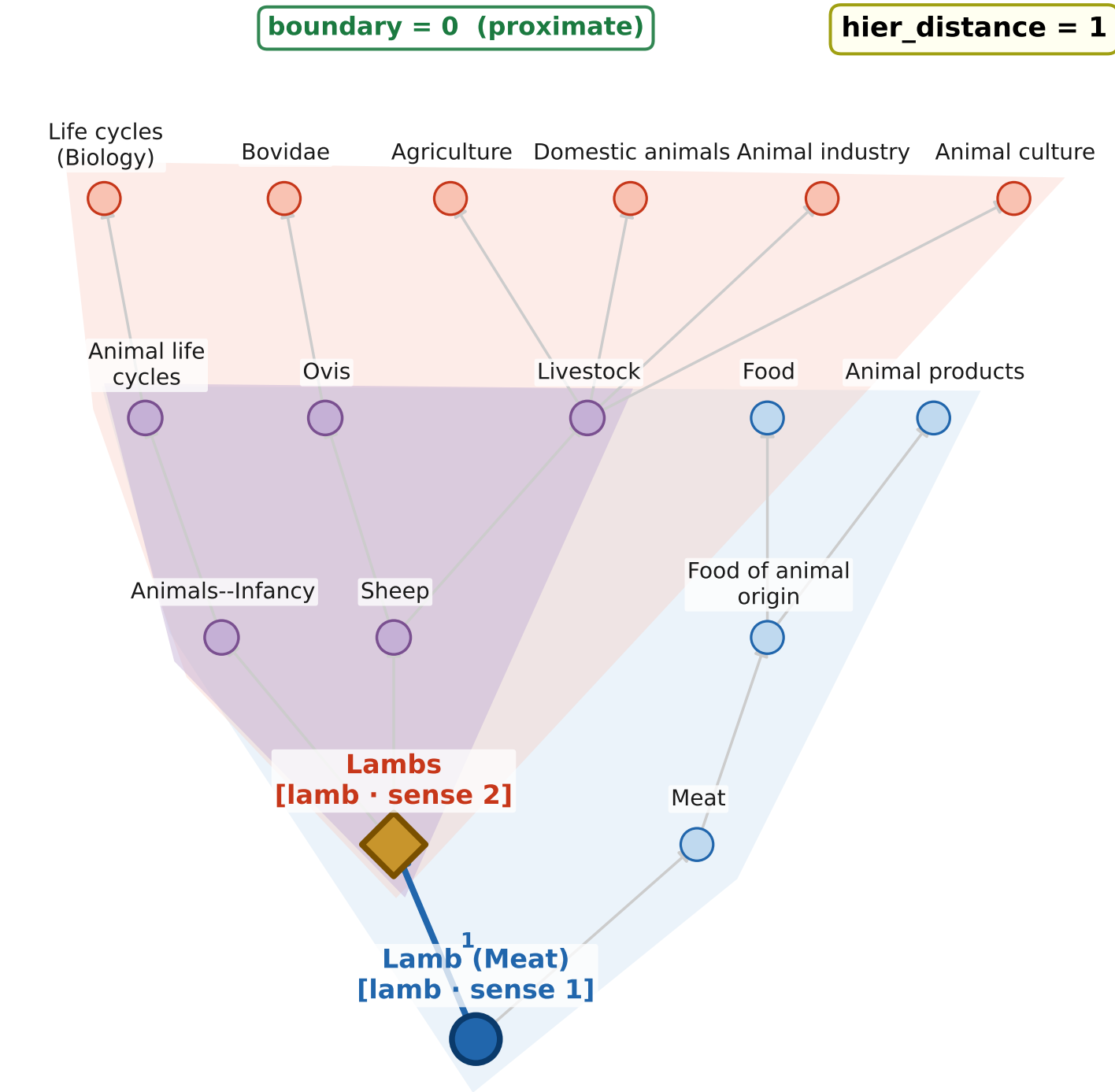
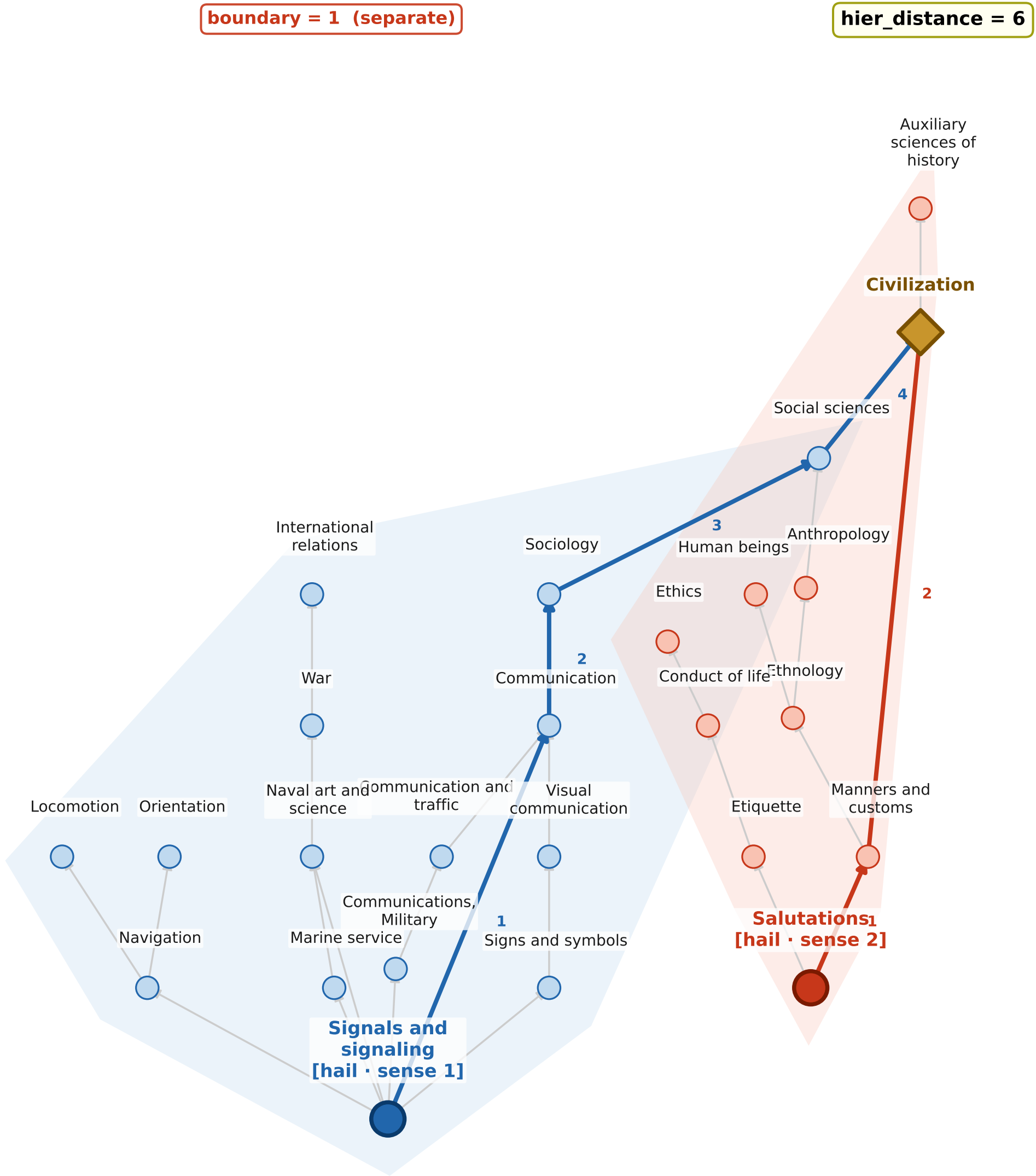


LCSH Hierarchical Subgraph Structure

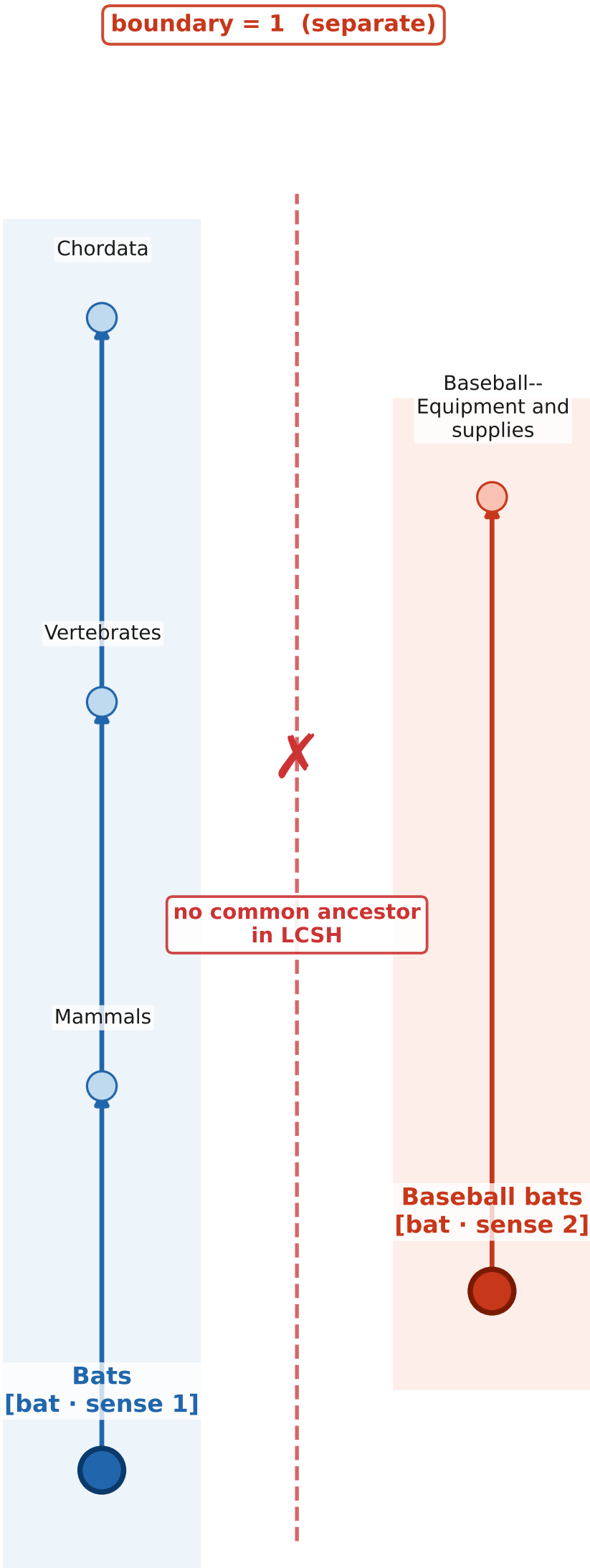
(A) lamb



(B) hail



(C) bat



Sense 1 heading node **Sense 2 heading node** **LCA node (◆)** **D≤3 neighborhood — sense 2** **D≤3 neighborhood — sense 1** **D≤3 overlap (both senses)** **Spine path — sense 1** **Spine path — sense 2** **Disconnected chains (no LCA)**

LCSH hierarchical subgraph structure for three polysemous words. Each panel shows a directed acyclic graph in which edges point from narrower (child) to broader (parent) term. Colored spines trace the route from each sense heading to their Lowest Common Ancestor (LCA, ◆ gold diamond); edge labels show cumulative hop distance from the sense heading. Shaded regions show the D≤3 neighborhood of each sense. hier_distance = total edges separating the two senses via their LCA. lcsch_boundary_binary: 0 = same hierarchical branch (proximate); 1 = separate branches or no common ancestor.